Homework Bootstrap, CLT, and Normal Approximation

1. $SE = \frac{10}{\sqrt{100}} = \boxed{1}$

$Z\text{-stat} = \frac{52 - 50}{1} = 2$

α		
0.1	1.645	> 2 reject null
0.05	1.96	> 2 reject null
0.01	2.576	> 2 fail to reject null

95% confidence interval

$(52 - 1(1.96),\ 52 + 1(1.96))$

$(50.04,\ 53.96)$

2. 90% CI $(3.1 - (1.3)(1.645),\ 3.1 + (1.3)(1.645)) = (0.9615, 5.2385)$

95% CI $(3.1 - (1.3)(1.96),\ 3.1 + (1.3)(1.96)) = (0.552, 5.648)$

99% CI $(3.1 - (1.3)(2.576),\ 3.1 + (1.3)(2.576)) = (-0.2488, 6.4488)$

0 is in interval

$Z\text{-stat} = \frac{3.1 - 0}{1.3} = 2.38$

$\alpha = 0.1$	$1.655 < 2.38$	reject null
$\alpha = 0.05$	$1.96 < 2.38$	reject null
$\alpha = 0.01$	$2.576 > 2.38$	fail to reject null of $S=0$

3.

$\mu = 12$

$\bar{x} = 11.5 \quad s = 3 \quad n = 36$

* Before doing anything, I believe 12 is a plausible value for the population mean.

95% CI $\left(11.5 - (1.96)\left(\frac{3}{\sqrt{36}}\right),\ 11.5 + (1.96)\left(\frac{3}{\sqrt{36}}\right)\right) = (10.52,\ 12.48)$

12 is in the interval

so we fail to reject null that $\mu = 12$

The conclusion suggests that 12 is a plausible value.

4.

$$Z = \frac{25-22}{\sqrt{\frac{8^2}{100} + \frac{7^2}{100}}} = \frac{3}{1.06} = 2.82 > 1.96$$

reject null

95% CI =

$$\left((25-22) - 1.96(1.06),\ (25-22) + 1.96(1.06)\right)$$

$$(0.916, 5.08)$$

5.

test statistic: $\frac{14-12}{\sqrt{\frac{6^2}{81} + \frac{5^2}{64}}} \approx 2.189$

$SE \approx 0.914$

6.

test statistic: $\frac{3.2-2.8}{\sqrt{\frac{1.5^2}{49}+\frac{1.4^2}{64}}} = \frac{0.4}{0.277} = 1.446$

$\alpha = .05$

$1.96 > 1.446$

CI 95% $= ((3.2-2.8)-(1.96)(0.277), (3.2-2.8)+(1.96)(0.277))$

$(-0.142, 0.942)$

The difference is not statistically large relative to the noise. We fail to reject the null at $\alpha = .05$.